Finding a geodesic current

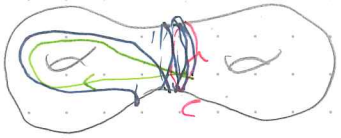
Idea Instead of looking at dense geodesics, work with closed geodesics. Need to deal with weights on the boundary of the intersting rectangles. (possible)

So pick an oriented simple closed curve c_0 on S

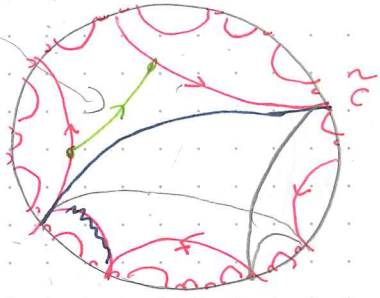
Look at all lifts to $\mathbb{H}^2$
(inital) Endpoints are dense.

Q separating or no

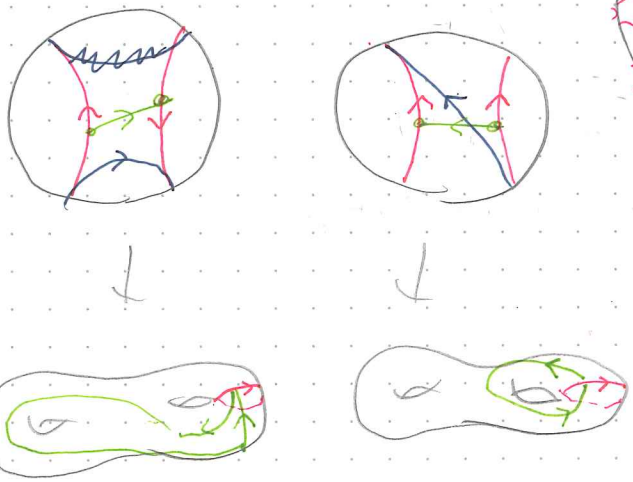
Suffices to find measure of flow boxes w/ endpoints at ends of lifts of c_0.
(Q open or closed?)



Have group elts relating these geodesics



Two cases:

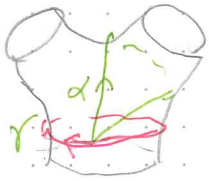


Depend on which side you approach c from

- Do both cases give a well-defined limit in toy model?
- can I restrict to endpoints of one of these types (and still get density)? Probably

compare 2018-09-06 #2

Toy example



$v([\alpha\gamma^n]) \geq n v(\gamma) + v(\alpha)$

(depends on some Dehn twist information)



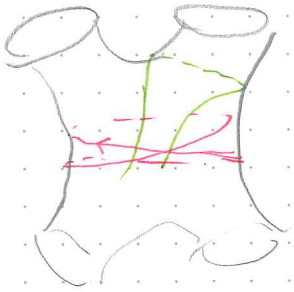
want to show

$\lim_{n\to\infty} v(\alpha\gamma^n) - v(\gamma^n)$ exists

Have ✓ lower bound on $v(\alpha\gamma^n)$

✓ upper bound?

convexity as fcn of n?

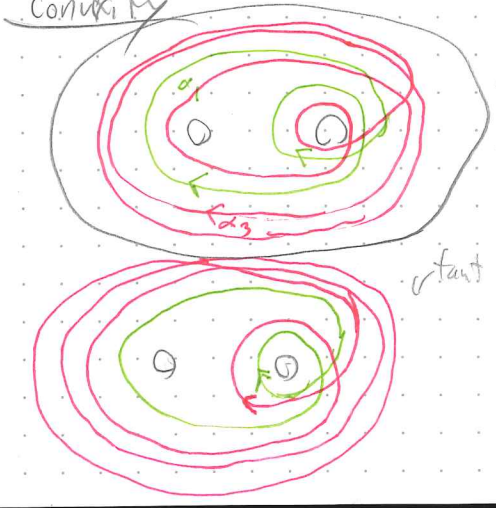




$v(\gamma^n) + v(x) \geq v(\alpha\gamma^{n-1}) + v(\beta)$

Also $v(\gamma^n) + v(x) \geq v(\alpha\gamma^{n-k}) + v(\beta\gamma^{k-1})$ FWIW

Convexity



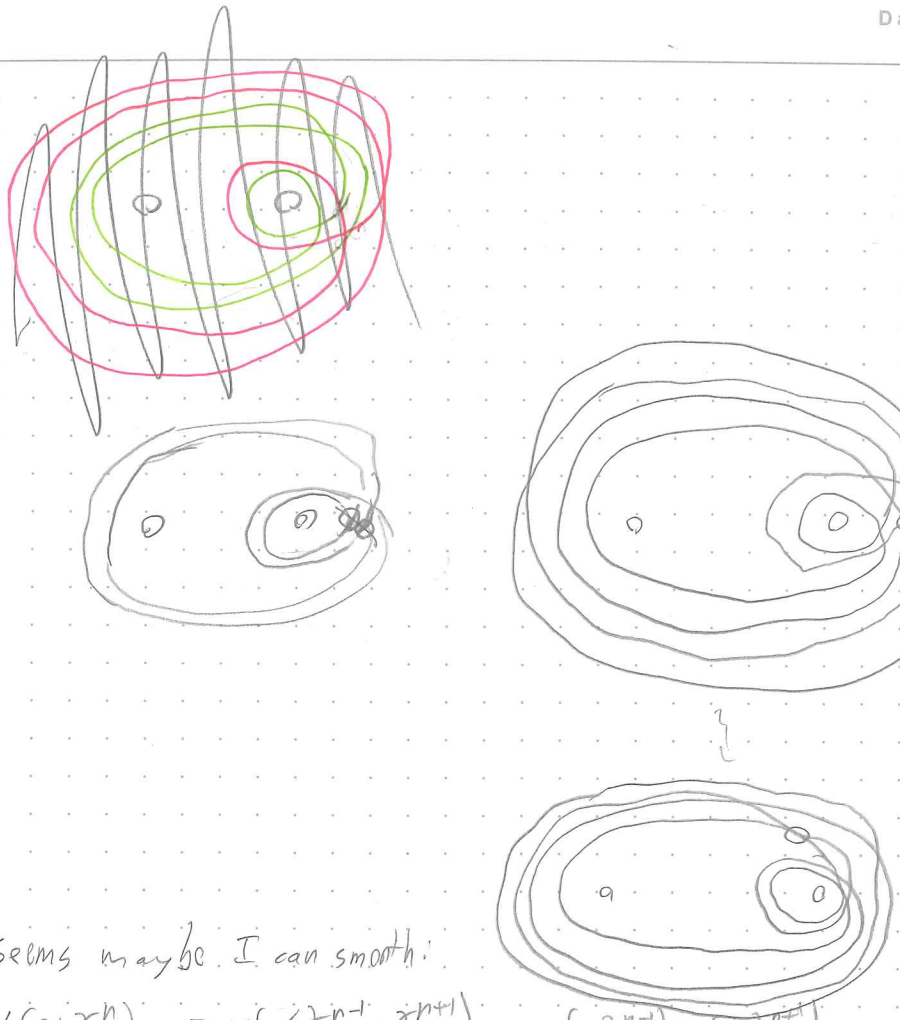
↙ Doesn't seem obvious that convexity is true

↖ not taut

↙ taut

smooths to $v(\alpha^2\gamma^4)$

Not $2v(\alpha\gamma^2)$



Seems maybe I can smooth.

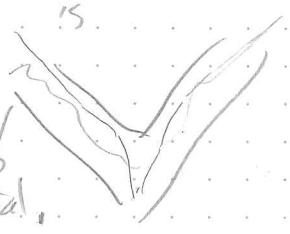
$2v(\alpha\gamma^n) \rightsquigarrow v(\alpha\gamma^{n-1}, \alpha\gamma^{n+1}) \rightsquigarrow v(\alpha\gamma^{n-1}), v(\alpha\gamma^{n+1})$

$\Rightarrow$ concave for n

Need to check this. How does it generalise?

Also, cannot globally be $*$ concave for n!

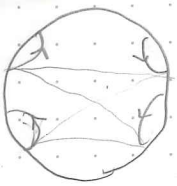
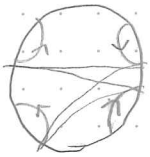
behaviour is



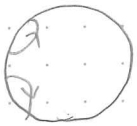
So cases w/ n near 0 must be special.

Note this kind of argument is more delicate - quasi-additivity is not enough.

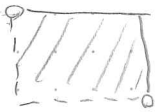
Overall picture several possibilities in a quadrilateral:



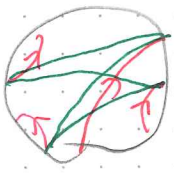
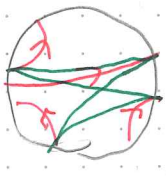





$[a,b] \times [c,d]$?


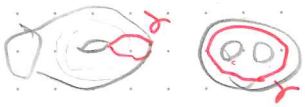
What can there actually be non-zero weights on the boundary of a long chamber?

- There are more cases than about 2^4 cases. Eg

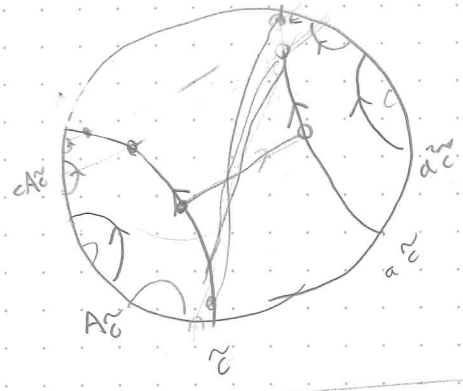
Is this one problematic?

- What cases for α, γ do we need to consider? (intersecting and essential intersections)
  - Pass to cover $\tilde{S}$ corresponding to $\langle \alpha, \gamma \rangle$?
  - Rank 2 subgroup $\Rightarrow$ surface $\tilde{S}$ is pair of pants or punctured torus
  - γ is simple $\Rightarrow$ put in standard form.
  - What can α be?



Is it just $\rho^2 \gamma \beta^{-1}$ — for standard dual generators? Structure of reducible automorphisms of F_2 —

What, exactly, do we measure w/ $\lim_{n\to\infty} v(\alpha\gamma^n) - v(\gamma^n)$?



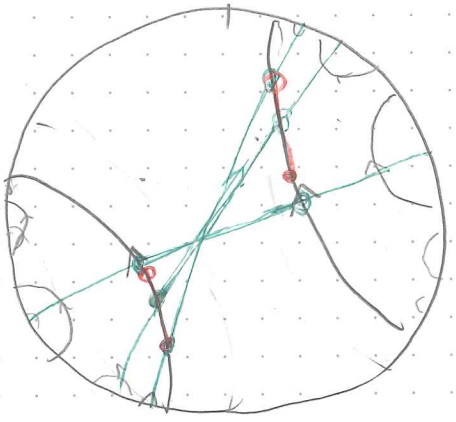
Outer space for F_2: Farey tiling + flaps

Kup, this claim works



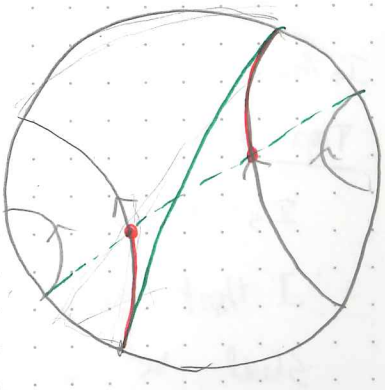
reducible $\Leftrightarrow$ fixes this point

Hmm: seems $Out(F_2) = GL_2(\mathbb{Z})$ acting on $\mathbb{H}^2$?



approximation

Limit!



~~intersection~~ γ (intersections w/ green − intersections w/ red)

Is this finite? Yes

Didac points out:

Additivity. ISSAP

Have weighted curves C_k defined by (smeared) first return map

$a_k = v(C_k)$ $\quad a_{k,\ell} = v(C_k \cup C_\ell)$

convexity: $a_{k,\ell} \leq a_k + a_\ell$

? $a_{k,\ell} \stackrel{?}{\sim} a_k + a_\ell$

stability: ~~$a_{2k} \leq 2a_k$~~ (right?) (wait, no) $\quad a_{k,k} = 2a_k$

smoothing: $a_{k+\ell} \leq v(C_k \cup C_\ell \cup X) \leq a_{k,\ell} + K \leq a_k + a_\ell + K$

isn't this already enough?

$a_{k,\ell} \leq v(C_{k+\ell} \cup X) \leq a_{k+\ell} + K$ ← seems to yield this for continuity

---

More convexity

Suppose you have a weighted directed embedded graph on a surface S

maybe assume no null-homotopic cycles, or else work in some completion

$P(\gamma) = \sum_{c \in \gamma} (w(c))$

Is this log-convex?

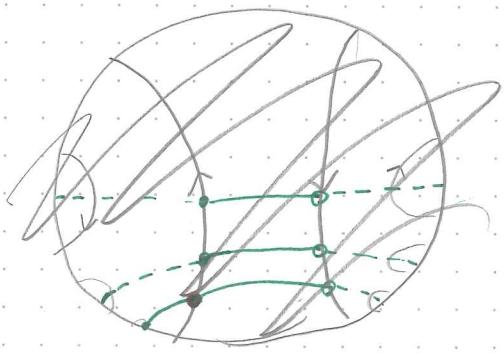
Tropicalise ⇒ length w.r.t. a metric

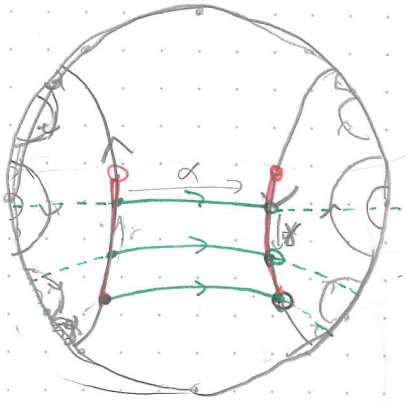
Is ∄ true w/o tropicalization?

I think so.

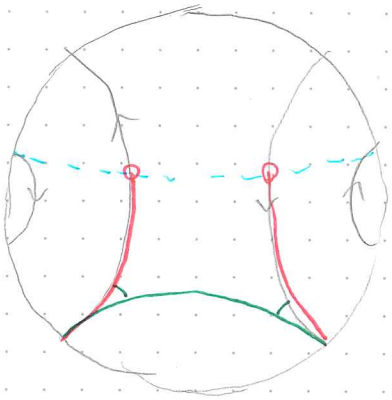
Should be some nice examples.

What is $\lim_{n\to\infty} v(\alpha\beta^n) - v(\beta^n)$ in pair-of-pants case?



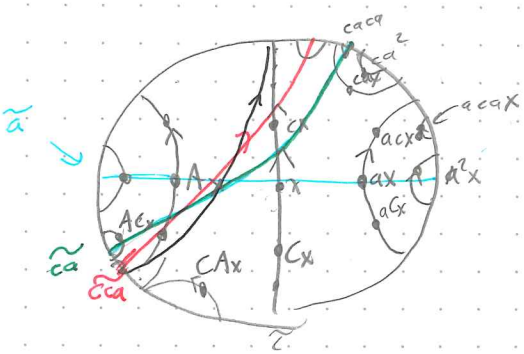


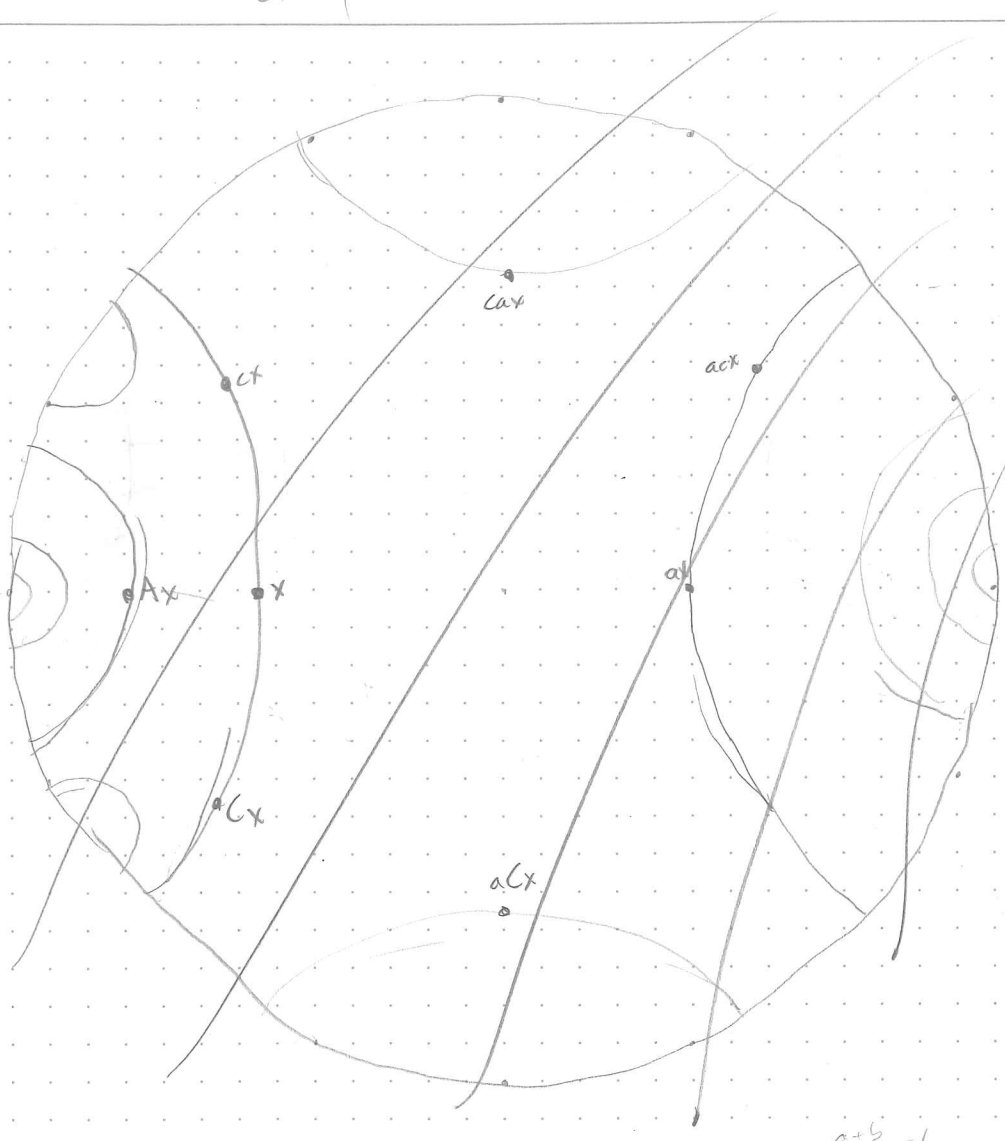
Limit



Something really funny.
Does not seem to work in combinations
Shouldn't these extra bits cancel out?

Ok, lets be more careful.





in $SL(2,\mathbb{R})$: $a(x) = 3x$

$$a(x) = \frac{3x+2}{2x+3}$$

$\frac{a+b}{c+d} = 1 \qquad \frac{a-b}{c-d} = -1$

$a+b = c+d$

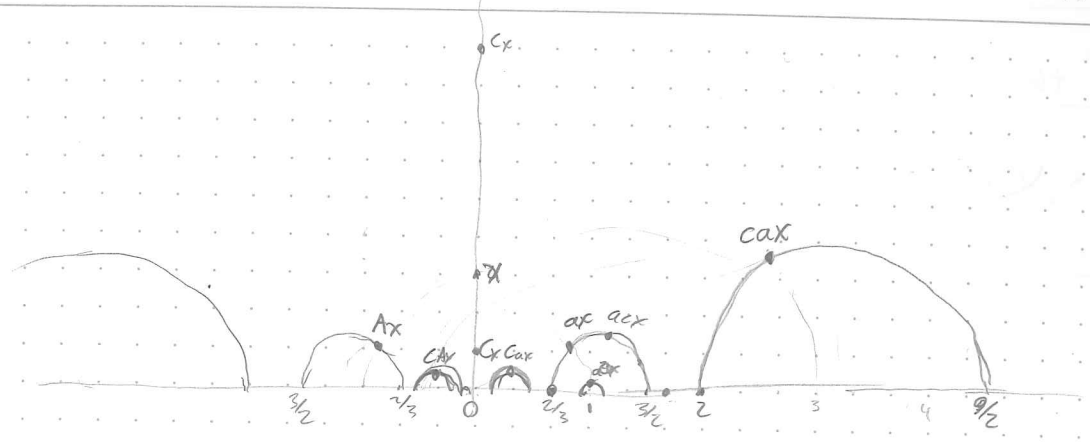
$a-b = d-c$

$a = d \qquad b = c$

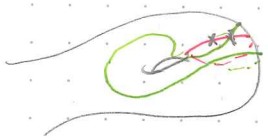
$$0, 1 \mapsto \frac{2}{3} \mapsto \frac{2+2}{4/3+3} = \frac{4}{13/3} = \frac{12}{13}$$

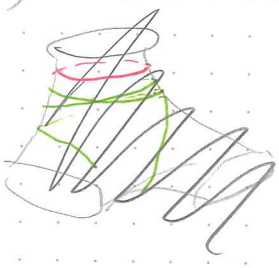
NB $x \mapsto kx$ is conjugate to $x \mapsto \frac{(k-1)x+k+1}{(k+1)x+(k-1)}$

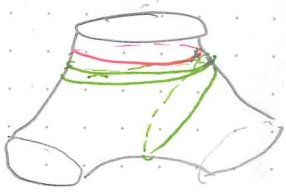
so $x \mapsto \frac{3x+2}{2x+3}$ is conjugate to $x \mapsto 5x$



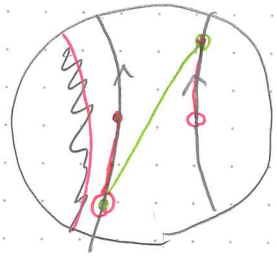
Simpler picture: cut on ξ, lift to $\zeta = 4^z$



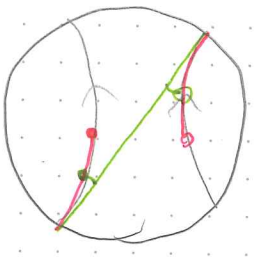




Torus



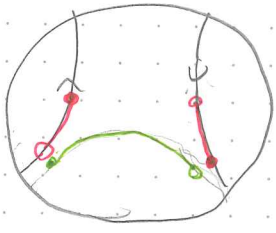
limit

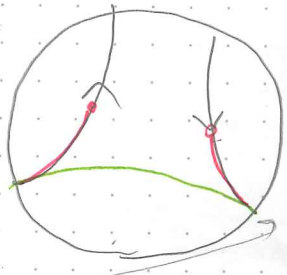


This is strange

Does seem to get very coherent answers
Am I doing the limits right?

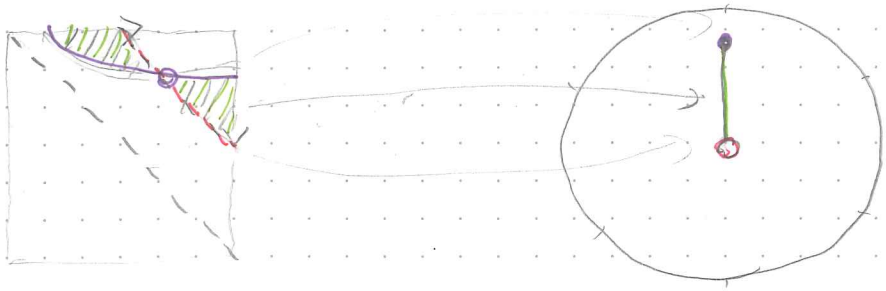
Pants

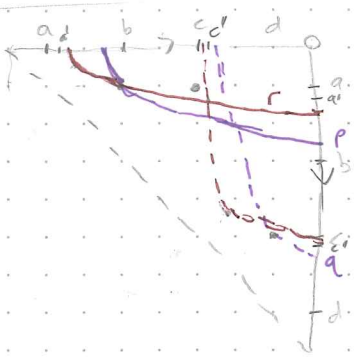


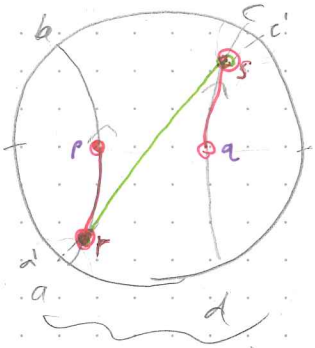


Don't include pencil

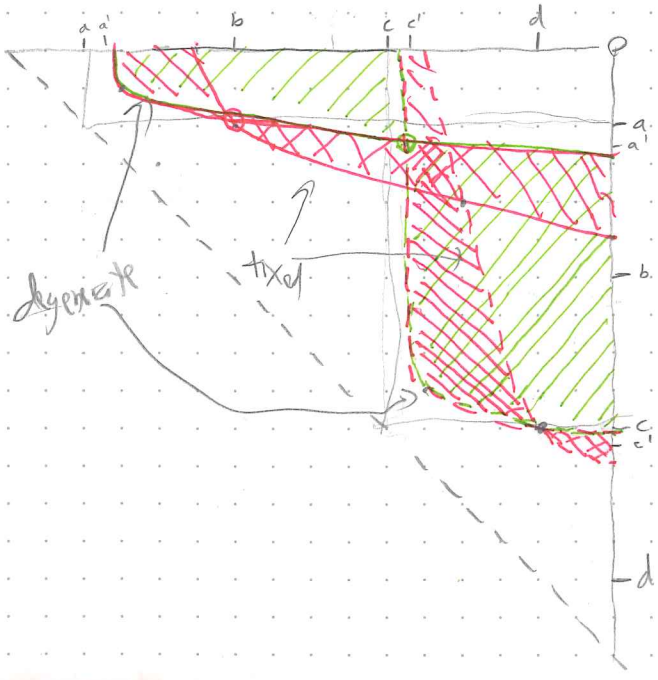
To get the limit right think about subsets of Möbius strip



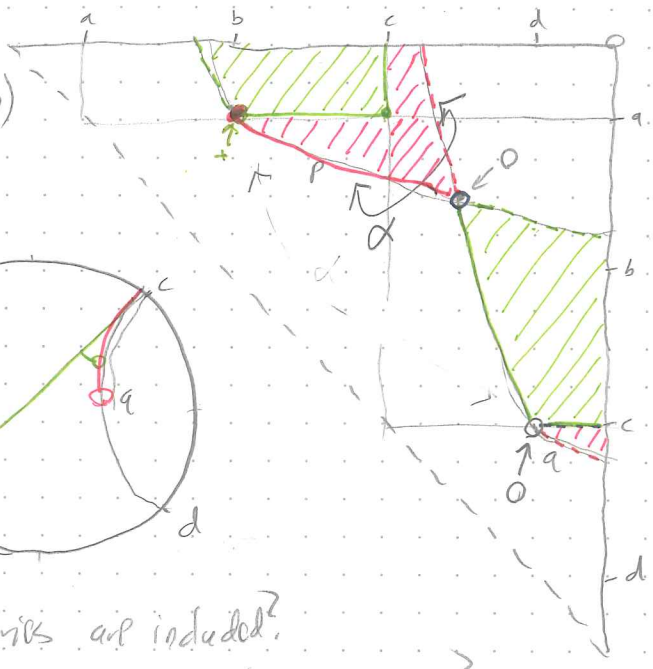




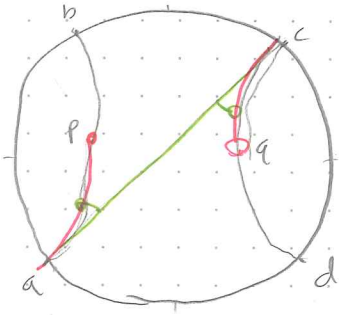
better switch around d to make covered region be positive



Limit (after cancellation)



Torus case

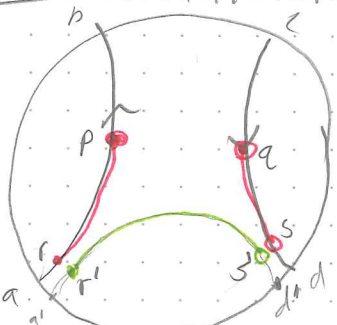


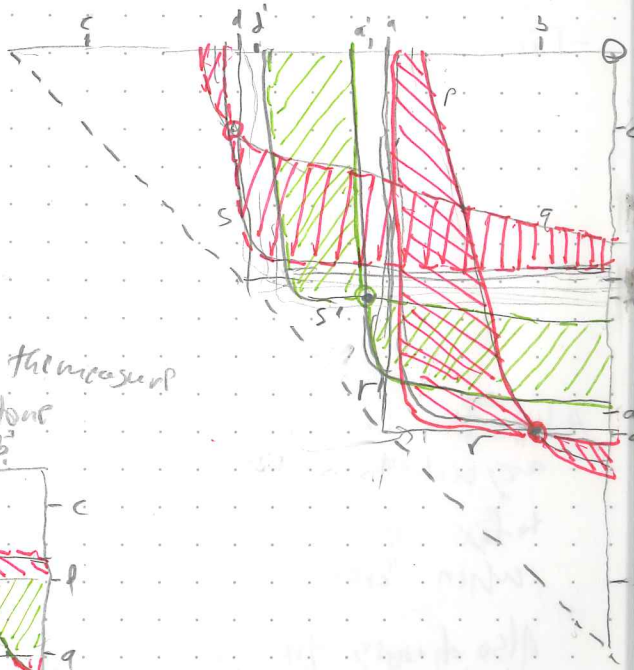
What boundaries are induced?

And do measures actually converge?

Yes converging from above (decreasing fn of n) ← positive region is decreasing, negative region is incr

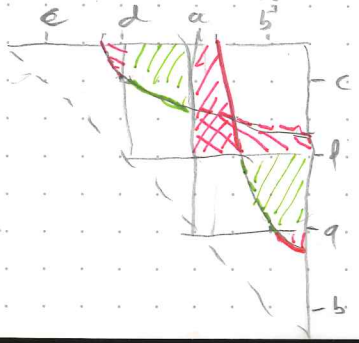
Pants case Seems like there will be a problem from lack of intersection





Issue: sets you are taking the measure of a–c not monotone

Limit



Page: 1

Date: 2018·09·22

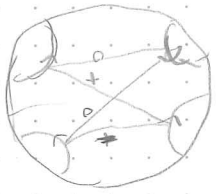
Ok, seems things are better. by Hopf argument (Alexandrov-Fenchel)

measure of the boundary is 0, except maybe for δ-masses on closed curves.

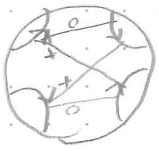
So just need to decide if these bdy pts are included or not.



questionable pts are way out of range. Not included in any counts anyway.

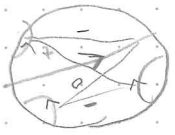


→ not counted



→ do count it!

Probably should but

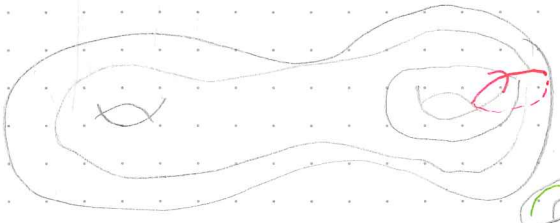


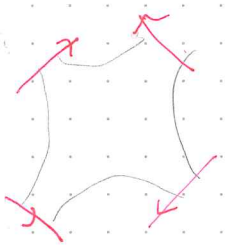
Ah, limit is negative?

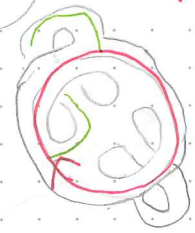
~~Counting is independent~~



- Forgot to check positivity at all. That's clear, smooth that central crossing...
- Finite additivity is really messy for naive rectangles. What does that even mean?





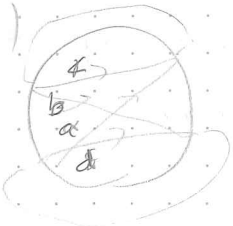


1 bdy
$\chi = -3$
genus 2

Ah hah — for smoothing positivity argument to work, need to fix the pairing. This changes when "crossing" happens.

Also changes the limit argument & makes it harder.

Pick (say)



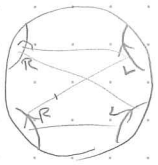
$a \mathbb{C} bD = id$

compute

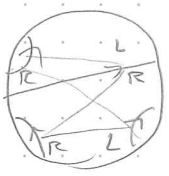
$f(ax^n bx^n) - f(cx^n) - f(dx^n)$

when does δ_x contribute?

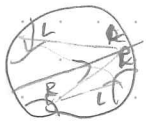
Eg



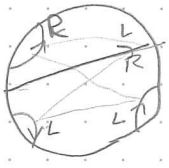
\+ : 2
− : 2
} net contributions clearly 0



\+ : 1 + 1
− : 2



diags: 2 + 1
sides: 1
} contributes



diags: 0 + 1
top/bot: 1

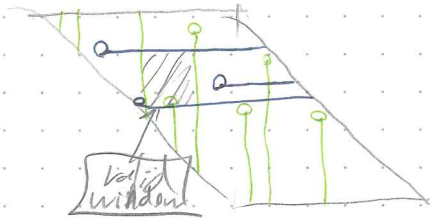
Getting messy. More annoying additivity brackets with this difference between

$v(a\cancel{a}^n b\cancel{x}^n)$ vs $v(ax^n) + v(bx^n)$

Idea
- Restrict which boxes are allowed to avoid "hard" cases
- Prove equivalence of multiple ways of finding the limit



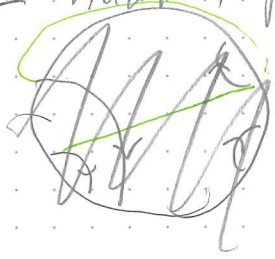
Each geodesic gives an "allowed region" for other endpts

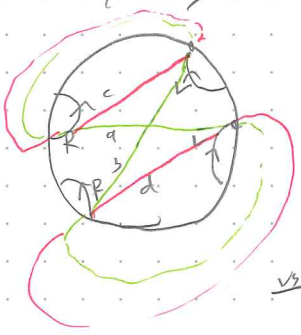


Such rectangles still cover the whole strip. I think.

seems likely that appropriate finite decomposition is true.

So have to prove equality of limits





vs the other one

$$\lim_{n\to\infty} f(ax^n bx^n) - f(ax^n) - f(dx^n)$$

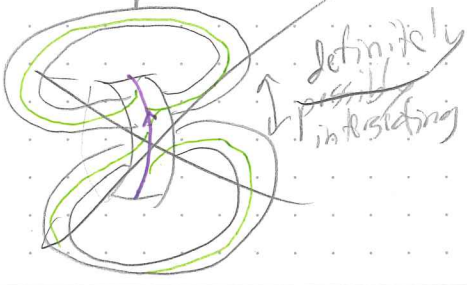
vs

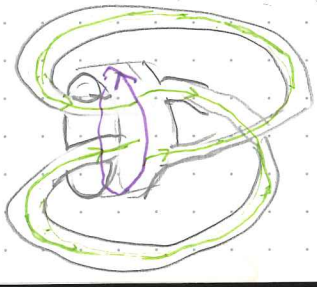
$$\lim_{n\to\infty} f(ax^n) + f(bx^n) - f(cx^n dx^n)$$

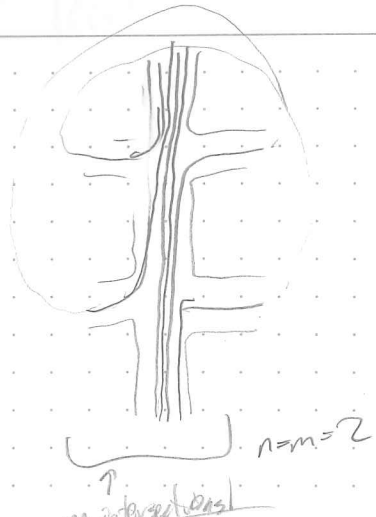
A posteriori, these limits are equal

Can I use some sort of convexity to prove that?

NB All set up in easier-to-deal-with setting, torus-like

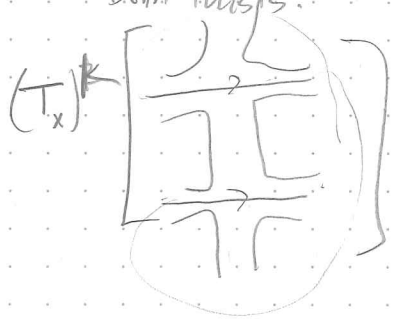






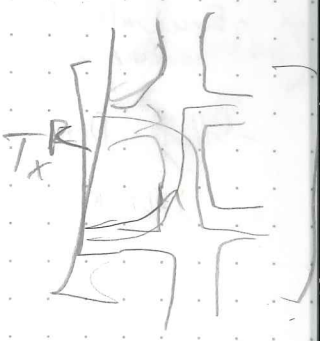
Oh—I should think about Dehn twists.

$(T_x)^k$

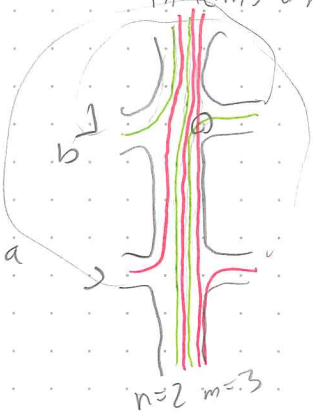


vs

T_x^k



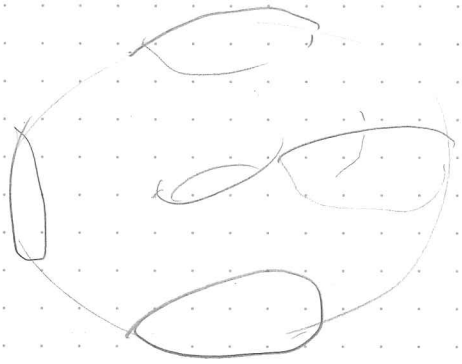
Should also be thinking in terms of joint limits: $\lim_{n,m\to\infty} f(ax^m bx^n) - f(cx^n) - f(dx^m)$



smooths to

$[ax^3 bx^2] \rightsquigarrow [bx^3] \cup [ax^2]$

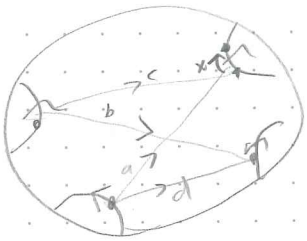
really depends on the details of how twisted a, b are



2018-10-01

What kind of convexity would help?

In setting

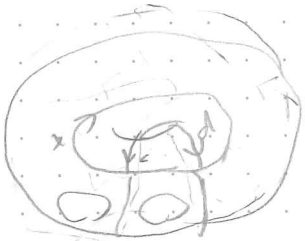


(probably dominated by worry about c/d)
(or maybe better: focus on c/d
minimal model on 2x punctured torus

have $f(n) = v(cx^n)$

$g(n) = v(dx^n)$

$h(n,m) = v(cx^n dx^m)$



know f, g, h change by at most $v(b)$ when a, g, s change

$|f(n) - f(m)| \le |n-m|\, v(x)$

$f(n+1) + f(n-1) \ge 2f(n)$ (convexity) $\Rightarrow$ limit exists

$f(n) + f(m) \ge |n-m|\, v(x)$

[NB: switching orientations]

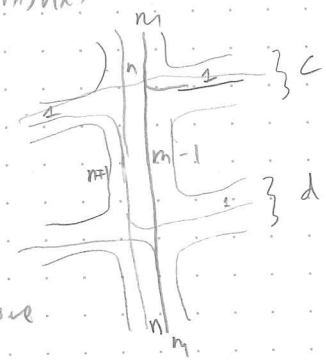
Ok, what about h?

$|h(n_1, m_1) - h(n_2, m_2)| \le (|n_1 - n_2| + |m_1 - m_2|)\, v(x)$

(n, m large, positive)

$h(n,m) + f(0) \ge g(n+m) \sim K_1 + (n+m)v(x)$

$h(n,m) + g(0) \ge f(n+m) \sim$



Know $f(n) \sim K_c + nv(x)$ ← approach from above

$g(n) \sim K_d + nv(x)$

Want $h(n,m) \sim K_c + K_d + (n+m)v(x)$

~~wait~~ ~~$h(n+k, m+k) + f(k) \ge g(n+m+k)$~~ ~~→ oops~~

$h(n,m) + f(k) \ge g(n+m-k)$ in a certain range ($n, m > k$?)

$\Rightarrow h(n,m) \ge K_c + K_d + (n+m)v(x) - \varepsilon$ ← this is nonsense

OK, but really want the other inequality too

Get that by sandwiching $[cx^n] \cup [dx^m]$ (as for convexity)

Need $n \ne m$ to actually have an intersection

Duh, duh $[c] \cup [dx] \searrow [cdx]$

$[cx^n] \cup [dx^{n+1}] \searrow [cx^n dx^{n+1}]$ oh, OK

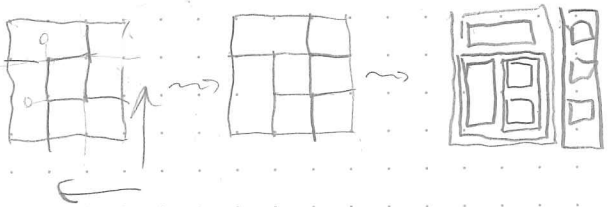
Page: 1

Date: 2018·10·02

Question 1 Is side-by-side additivity enough to prove finite additivity in whisker context?

Question 2 Can you pass to a cover where $c.d.$ both intersect $\mathbb{X}$ just once (so you can do twisting arguments?

Q1 Example:



Given $x, c, d \in \pi_1(S)$ x simple

Consider T = cover of S corresponding to $\langle x, c, d \rangle$

T is a rank 3 surface, if non degen

$\Rightarrow$  or 

In context know x is simple, non separating $\Rightarrow$ T is $T^2 \setminus$ (2 disks)

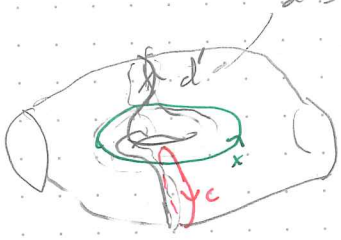
Make x longitude

Q Can we make c, d simple (after pre/post multiplying by powers of x)?

Recall/fixed a question about bases for F_3 with one generator

Is there an analogue for outer space here?

I don't think there's a "basis" thm along the same lines. Eg



not simple. Generates π_1, so have to work in cover.

---

oriented vs unoriented

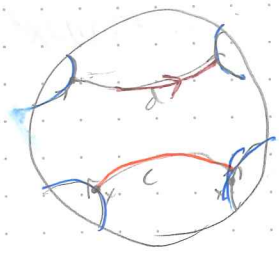
If $\vec{v}$ is an oriented convex fcn, what is $v(\vec{c}+\overleftarrow{c})$?



$v(D(c)) \searrow D(\text{smoothing of } c)$

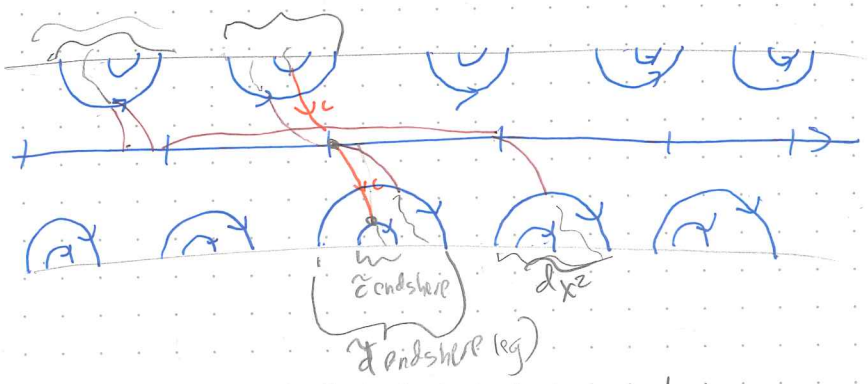
so $v(D(c))$ is convex

Remains to analyze other half $v(\vec{c}-\overleftarrow{c})$. Doesn't seem to be convex.

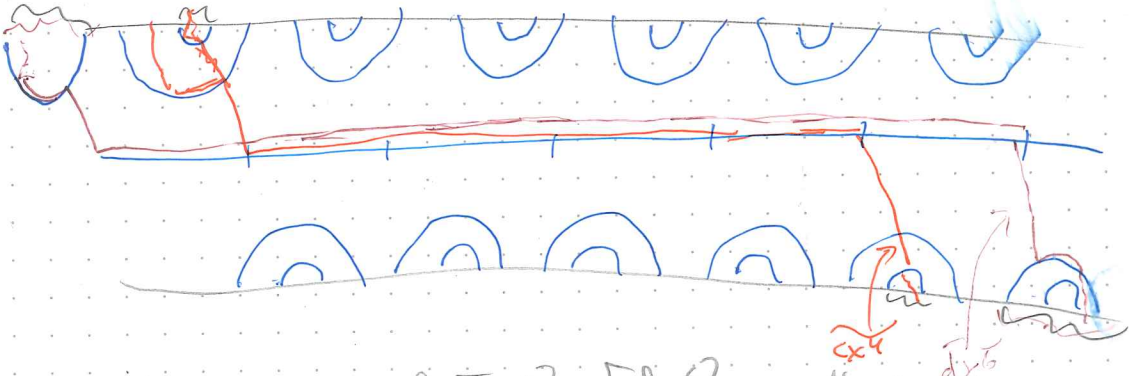


interested in intersections between cx^n & dx^m in nbhd of x.

center window on x
(band model)



dx^2 & c definitely cross: so do cx^n and dx^{n+2}



this union of $[cx^4] \cup [dx^6]$ smooths to

$[dx^5] \cup [cx^5]$

In general with this setup, can smooth

$[cx^n] \cup [dx^{n+2}] \rightarrow [cx^{n+1} dx^{n+1}]$

$[cx^n] \cup [dx^{n+3}] \rightarrow [cx^{n+1} dx^{n+2}]$
or
$[cx^{n+2} dx^{n+1}]$

$[cx^n] \cup [dx^{n+k}] \rightarrow [cx^{n+l} dx^{n+k-l}]$ for $1 \le l \le k-1$

(these cases will vary)

Cool. Looks like this will work for the other smoothing too.

Date: 2018 · 10 · 04

Q. Can I smooth $[cx^n dx^m] \cup [cx^k]$ to $z[cx^l] \cup [dx^r]$

for some exponents? (i.e., preserve orientations in lower bound on $h(n,m)$.

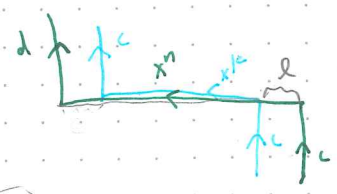
~~knowledge~~ $h(n,m) + f(k) \geq 2f(l) + g(r)$

~~Oops, not good enough. This gives (prob)~~

~~$h(n,m) \leq 1$~~

In fact that's what I need in an oriented case too.

$[cx^n \cdot dx^m] \cup [cx^k]$ focus on c.

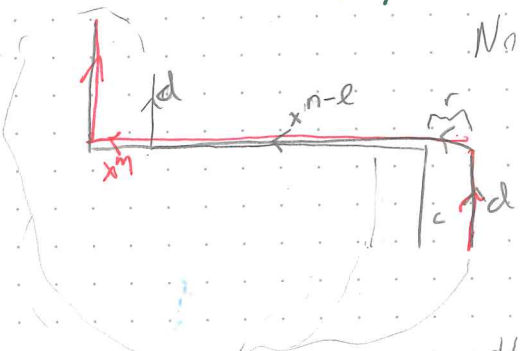


$k<n$ (& const) smooth to $[dx^m c x^{k+l} c x^{n-l}]$

w/ possible values of l depending on $n-k$

Now want to smooth that to

$[dx^{n-l+r}] \cup [cx^{k+l} cx^{m-r}]$



optional with $k+l=m-r$

[or maybe do so

[can also add in some extra pairs of c to get to exact double

Assume $n=m$, can smooth as on previous page

$[cx^n dx^n] \cup [cx^{n-4}] \searrow [dx^n cx^{n+2} cx^{n-2}] \searrow [dx^{n-1}]$ $[cx^{n+2} \cdot x^{n-1}]$

$[cx^n dx^n][cx^{n-4}][x^3] \searrow [dx^{n-1}][[cx^{n+2}]^2]$

Q What's an obstruction for a subdivision rule?
(why haven't I discussed this yet?)

several domains to ask for geometric subdivision rule
- conformal
- Euclidean similarities
- Affine

To prove A convex, additive curve functional comes from intersection #

① Pick a simple closed curve $[x]$, pick basept $p \in [x]$

② For any $c \in \pi_1(\Sigma, *)$ crossing X → define

$$\lim_{n\to\infty} v([cx^n]) - v([x^n])$$

exists.

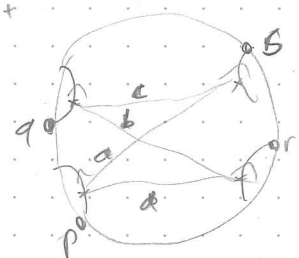
③ For $c, d \in \pi_1(\Sigma, *)$ crossing X (w/ same sign).

$$\lim_{m,n\to\infty} v([cx^n dx^m]) - v([cx^n][dx^m]) = 0$$

④ For $p, q, r, s \in \partial^\infty \tilde{\Sigma}$ suitably arranged

lifts of $\tilde{x}^-$ lifts of $\tilde{x}^+$

a, b, c, d translations relating lifts



set $\mu([p,q] \times [r,s])$

$$:= \lim_{n\to\infty} v([ax^n][bx^n]) - v([cx^n][dx^n])$$

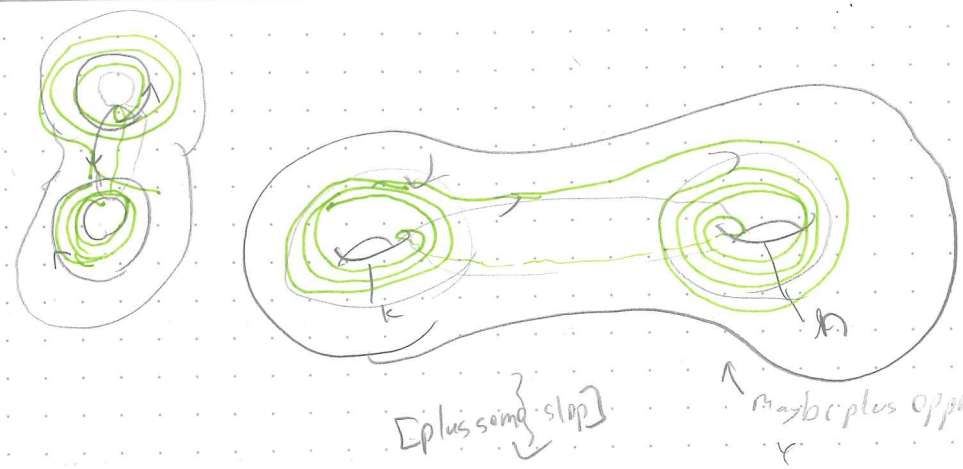
This exists, ≥ 0

⑤ "suitable rectangles" form a ring of sets

⑥

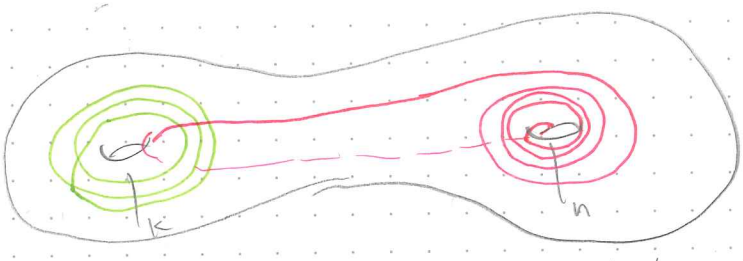
Page: 1

Date: 2018·10·07



[plus some slop]

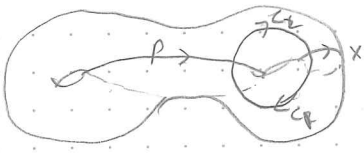
↑ maybe plus opposite spiral



Looks promising actually

To find $\nu(p)$ given the measure at ∞ (w/ winden construction

- Pick curve c crossing both p & X



~~Consider~~ wait

$$\frac{1}{2}\left([c_1 p^k c_2 x^n] + [c_1 p^k c_2 x^n]\right)$$

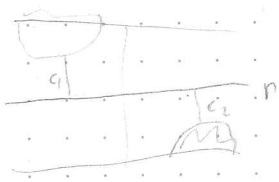
$$-[c x^n] \overset{?}{\sim} k[p]$$

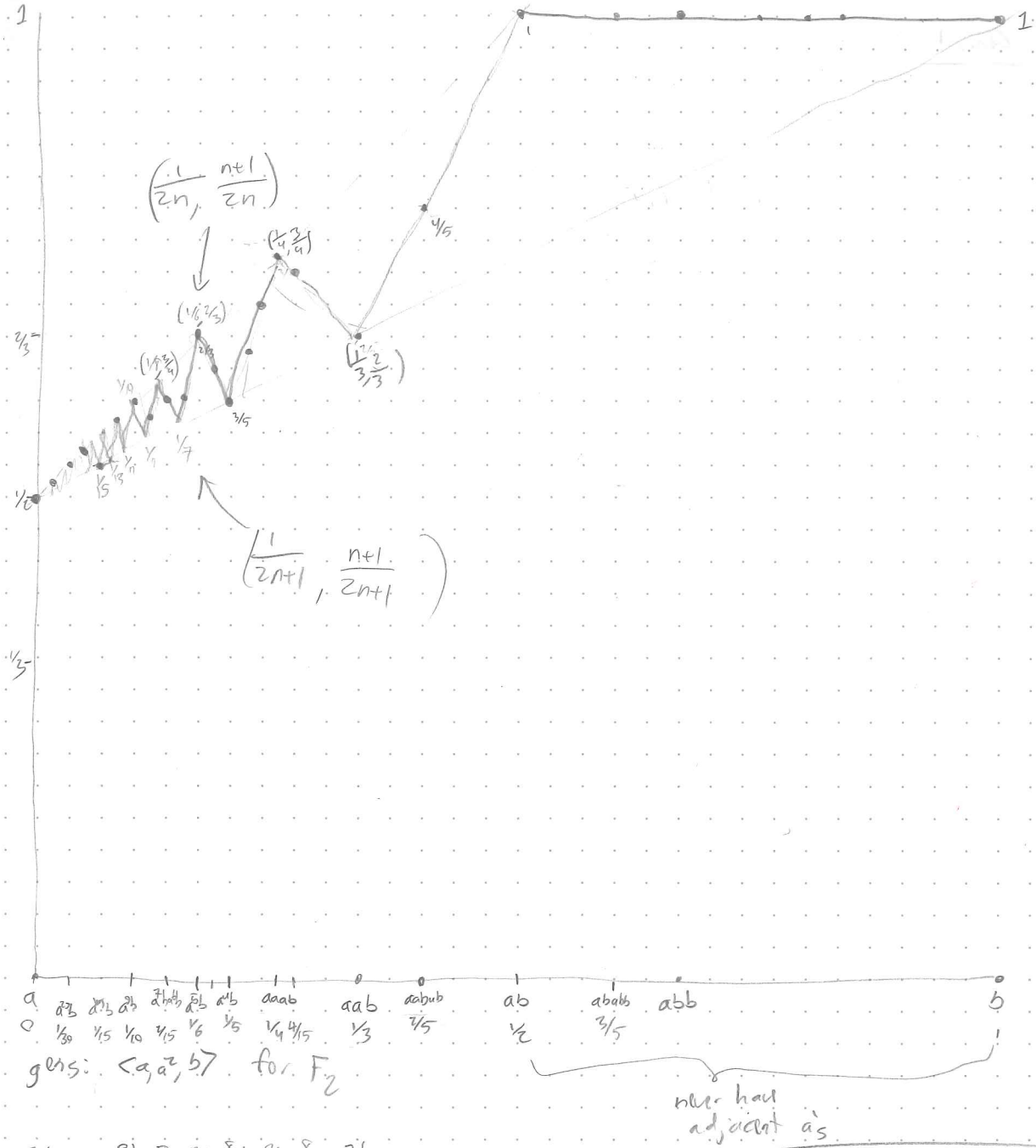


or better:

$$[c_1 p^k c_2 x^n] \sim k[p] + n[x]$$

$$[p^k]\ [x^n][c] \searrow [c x^n][p^k] \searrow [c_1 p^k c_2 x^n]$$





7/60: $a^8b\ a^8b\ a^7b\ a^8b\ a^7b\ a^8b\ a^7b$

3/20: $a^6b\ a^6b\ a^5b$

11/60: $a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^4b$

38/60

13/60: $13 \times b$'s $5x\ a^3\ 8x\ a^4$

7/30: $7 \times b$ $5x\ a^3\ 2x\ a^4$

4/15: $4 \times b$ $1x\ a^2\ 3x\ a^3$

Wait. This does not satisfy the bounds

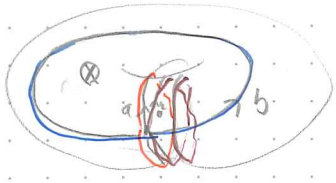
$f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2} + C|x-y|$

So where did argument go wrong?

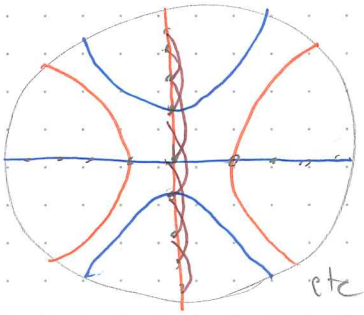
Concretely.

Gens:

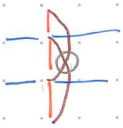
in S:

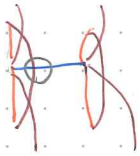


in $\tilde{S}$:



crossings in $\tilde{S}$:

(A)  $K(x) = 4-2 = 2$

(B)  $K(y) = 3-1 = 2$

Claim $\ell(c) \geq \ell(c') - 2$

c' a smoothing of c

Eg $\ell([a^2b][b]) = 3 \geq \ell([abab]) - 2 = 2$

$\ell([a^2]) = 1 \geq \ell([a][a]) - 2 = 0$

$\ell([a^2][a^4b]) = 5 \geq \ell([a^3ba^3b]) - 2 = 6-2 = 4$

Not getting equality?? That's fine, don't really care

~~$\ell([a^{12}b][a^{14}b]) =$~~

$f(\frac{1}{13}) = \overline{x}(\frac{1}{13}[a^{12}b]) = \frac{7}{13}$

$f(\frac{1}{15}) = \overline{x}(\frac{1}{15}[a^{14}b]) = \frac{8}{15}$

$f(\frac{1}{14}) = \overline{x}(\frac{1}{14}[a^{13}b]) = \frac{4}{7} = \frac{8}{14}$

$i(\frac{1}{13}[a^{12}b], \frac{1}{15}[a^{14}b]) = \frac{2}{13\cdot 15}$ → really small

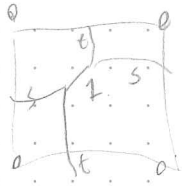
$\frac{1}{14} = (\frac{13}{28})\frac{1}{13} + (\frac{15}{28})\frac{1}{15}$

$f(\frac{1}{14}) = \frac{1}{28}\ell([a^{13}b])$

$= \frac{1}{28}(\ell([a^{12}b]) + \ell([a^{14}b]) + 1)$

$= \frac{13}{28} f(\frac{1}{13}) + \frac{15}{28} f(\frac{1}{15}) + \frac{1}{28}$

↓ way bigger than $i(\frac{1}{13}[a^{12}b], \frac{1}{15}[a^{14}b])$

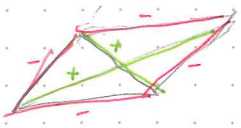


$s+t=1$

Parallelogram law

why does parallelogram law imply additivity?

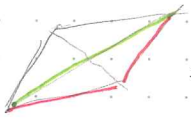
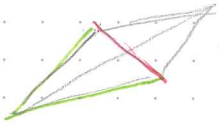
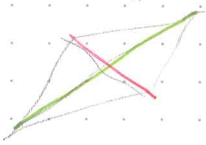
parallelogram law (combs of $\|\cdot\|^2$)

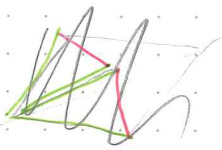


$= 0$

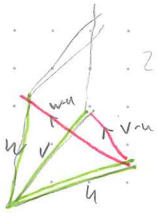
$u+v$
$u-v$ — u
v

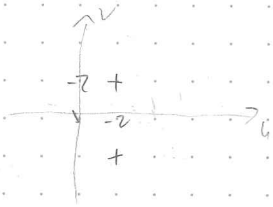
Define inner product

$2\langle u, v\rangle =$  $=$  $= \frac{1}{2}$ 

$2(\langle u, v\rangle + \langle u, w\rangle)$ 

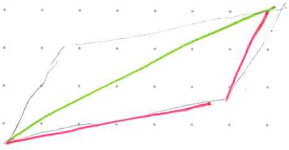




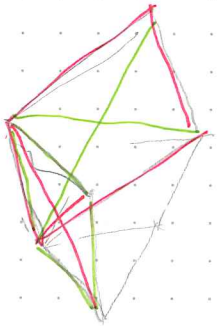


$= \|u+v\|^2 - \|u\|^2 - \|v\|^2 + \|u+w\|^2 - \|u\|^2 - \|w\|^2$

$\|u+v\|^2 - \|u-v\|^2 + \|u+w\|^2 - \|u-w\|^2$

$\langle u, 2v\rangle =$ 

Parallelogram
$(u+w, v)$
minus $(u-w, v)$

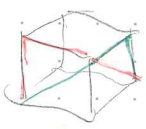


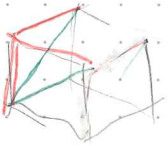
$\|u+w+v\|^2 + \|u+w-v\|^2 = 2\|u+v\|^2 + 2\|w\|^2$
$\|u+w+v\|^2 + \|u-w-v\|^2 = 2\|u-v\|^2 + 2\|w\|^2$

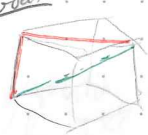
Not really reasonable.

OK, just punt.

Goal







I think these inequalities were going the wrong way, no?

$EL = \sup \frac{\ell^2}{A}$ want $\sqrt{EL[c]} \geq \sqrt{EL[c']} - K$

Need to start w/ an optimizer for c', not for c



~~Soon~~ OK, we're good:

Pick optimizer for c': $\sqrt{EL[c']} = \frac{\ell_p[c']}{\sqrt{A_p}} = \ell_p[c']$

Normalize $A_p = 1$

$$\sqrt{EL[c]} \geq \ell_p[c] \geq \ell_p[c'] - NK$$

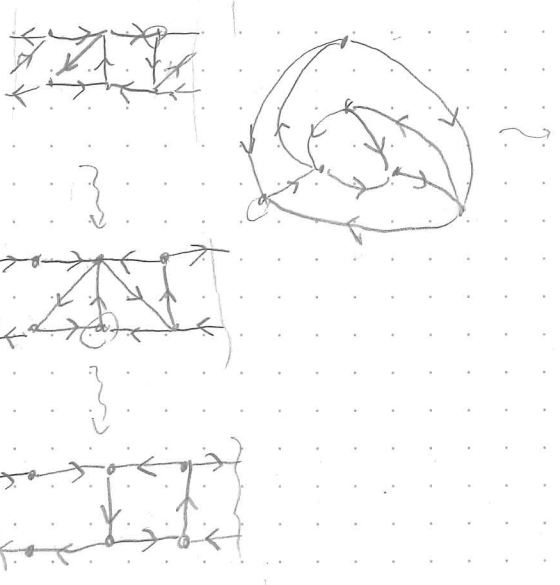
What about $SL(3)$ positivity?

Key case: Annulus

this cluster alg.



(rank 6)



What are the imaginary roots of this cluster algebra?

Wait, back up. In SL_2 case, the algebraic thing is $\operatorname{tr} A = \lambda + \lambda^{-1}$

+ Take log to make additive instead of multiplicative

- Does not satisfy smoothing; things are sign-definite, not always positive

- stable version is $\lim_{n\to\infty} \frac{1}{n} \log(\operatorname{tr} A^n) = \lambda$

- For $SL(3)$, similarly get $\lim(\quad) = \lambda_1$ $\quad \left(\lambda_1, \lambda_2, \lambda_3\right)$ $\pi = 1$

Ok! So that is the right characteristic fcn

Now Does this satisfy the smoothing inequality?

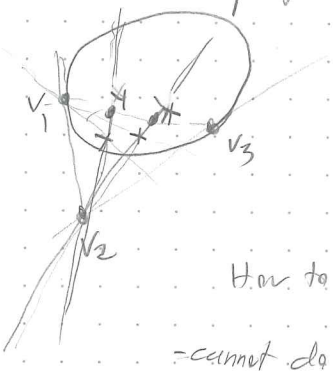
What's the class of reps. we are looking at?

"anosov" → eg: int map $\partial\widetilde{\Sigma} \to \mathbb{RP}^n$ (or better to flags in $\mathbb{RP}^n$)
or connected component of image of Fuchsian reps

$\widetilde{\pi}_1(\Sigma) \to SL(2) \to SL(n)$

To prove Maybe pass to cover generated by crossing?

For $SL(3)$ Convex projective structures $\Rightarrow$ get an asymmetric metric?



Matrix: $A(p v_1 + q v_2 + r v_3) = p\lambda_1 v_1 + q\lambda_2 v_2 + r\lambda_3 v_3$

For a geodesic $x = \{s v_1 + t v_3\}$

$y = \{s\lambda_1 v_1 + t\lambda_3 v_3\}$

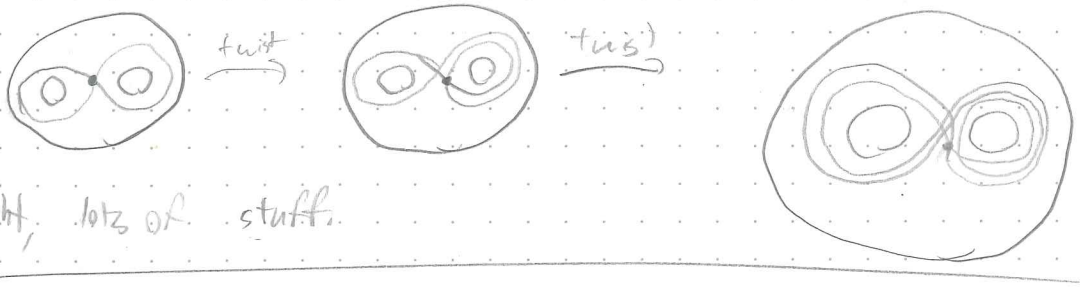
How to extract λ_1? (not λ_1/λ_3)

- cannot do it just from behaviour on $v_1 - v_3$ line (projectively)

Date: 2008 · 10 · 15

Intersections lifted to maximal cover

I think there are many pair-of-pants possibilities:

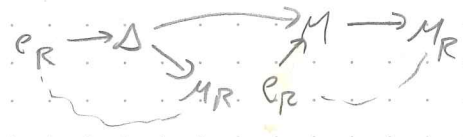


Right, lots of stuff.

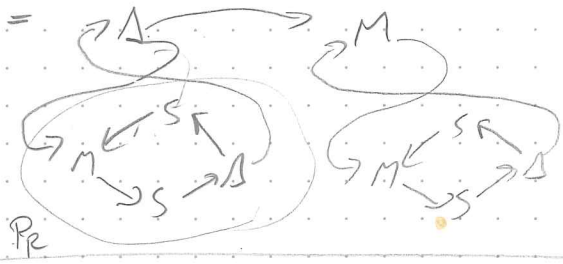
Greg says

Framing change elt z

$a \to \alpha$ =



=



What is this concretely? I worked out P_R before, but lost it...

conventions 2008-09-13 #2

P_R

Δ

